LAB#5

URL APIs, navigation, multiple views, and JSON parsing. Sample code is provided and you must finish the LAB as instructed by the professor.

Email swiftUI, Swift programming code and a screenshot of the simulator display to Kevin Kairouz, the TA for this class. His email will be given in the Bb -> Announcements. Thank you.

Add three (3) more JSON URLs navigation links for this lab assignment. You can pick any three that you want. Do the same as the “currency” and “todos” JSON lists.

Xcode playground Swift code:

```swift
import UIKit

let json = """
{
    "flight_number": 65,
    "mission_name": "Telstar 19V",
    "launch_date_unix": 1532238600,
    "launch_success": true
}
"""

struct Launch: Decodable {
    let flightNumber: Int
    let missionName: String
    let launchDateUnix: Date
    let launchSuccess: Bool
}


let jsonData = json.data(using: .utf8)!
let decoder = JSONDecoder()
decoder.keyDecodingStrategy = .convertFromSnakeCase
decoder.dateDecodingStrategy = .secondsSince1970
let launch = try decoder.decode(Launch.self, from: jsonData)

print(launch.flightNumber)
print(launch.missionName)
print(launch.launchDateUnix)
print(launch.launchSuccess)
```

```swift
//
//  ContentView.swift
//  lab#4MobileProject
//
//  Created by Nick Kounavelis on 9/28/22.
//

import SwiftUI

struct ContentView: View {

    var body: some View {
        NavigationView {

            VStack {
                // 1) JSON URL  ------------
                NavigationLink(destination: CurrencyView()) {
                     Text("World Currency Abbreviations")
                        .font(.title3)
                        .padding(.bottom, 15)
                }


                // 2) JSON URL  ------------
                NavigationLink(destination: TodosView()) {
                     Text("TO DO LIST")
                        .font(.title3)
                        .padding(.bottom, 15)
                }

            }  // end vertical Stack
            .navigationTitle("JSON SAMPLE URLs")

        } // end navigation view

    } // end body

} // end struct

struct ContentView_Previews: PreviewProvider {
    static var previews: some View {
```

```
        ContentView()
    }



//
//  CurrencyView.swift
//  lab#4MobileProject
//
//  Created by Nick Kounavelis on 9/28/22.
//

import SwiftUI

struct Currency: Decodable {
    let id: String
    let name: String
    let min_size: String
}

struct CurrencyList: Decodable {
    let data: [Currency]
}

struct CurrencyView: View {
    @State private var currencies = [Currency]()

    var body: some View {
        NavigationView {
            List(currencies, id: \.id) { c in
                VStack (alignment: .leading) {
                    Text(c.id)
                        .font(.headline)
                        .foregroundColor(.cyan)
                    Text(c.name)
                        .font(.body)
                        .foregroundColor(.indigo)
                    Text(c.min_size)
                        .font(.body)
                         .foregroundColor(.red)
                }
            }
            .navigationTitle("World Currency")
            .task {
                await fetchCurrencyData()
            }
        }
    }
```

```
    func fetchCurrencyData() async {
        // create the URL
        guard let url = URL(string:
"https://api.coinbase.com/v2/currencies") else {
            print("Hey Man, THIS URL DOES NOT WORK!")
            return
        }

        // fetch the data
        do {
            let (data, _) = try await URLSession.shared.data(from: url)

            // decode that data
            if let decodedResponse = try?
JSONDecoder().decode(CurrencyList.self, from: data) {
                currencies = decodedResponse.data
            }
        } catch {
            print("Bad news ... This data is not valid :-(")
        }

        // ecode the data
    }
}

struct CurrencyView_Previews: PreviewProvider {
    static var previews: some View {
        CurrencyView()
    }
}
```

```swift
//
//  TodosView.swift
//  lab#4MobileProject
//
//  Created by Nick Kounavelis on 9/28/22.
//

import SwiftUI

struct Todo: Decodable {
    var userId:      Int
    var id:          Int
    var title:       String
    var completed:   Bool
}

struct TodosView: View {
    @State private var todos = [Todo]()

    var body: some View {
        NavigationView {
            List(todos, id: \.id) { t in
                VStack (alignment: .leading) {
                    Text(String(t.id))
                        .font(.headline)
                        .foregroundColor(.cyan)
                    Text(t.title)
                        .font(.body)
                        .foregroundColor(.indigo)
                    Text(String(t.completed))
                        .font(.body)
                        .foregroundColor(.red)
                }
            }
            .navigationTitle("Todo List")
            .task {
                await fetchTodoData()
            }
        }
    }


    func fetchTodoData() async {
        // create the URL
```

```
        guard let url = URL(string:
"https://jsonplaceholder.typicode.com/todos") else {
            print("Hey Man, THIS URL DOES NOT WORK!")
            return
        }

        // fetch the data
        do {
            let (data, _) = try await URLSession.shared.data(from:
url)

            // decode that data
            if let decodedResponse = try?
JSONDecoder().decode([Todo].self, from: data) {
                todos = decodedResponse
            }
        } catch {
            print("Bad news ... This data is not valid :-(")
        }

        // ecode the data
    }
}


struct TodosView_Previews: PreviewProvider {
    static var previews: some View {
        TodosView()
    }
}
```

{"data":[{"id":"AED","name":"United Arab Emirates Dirham","min_size":"0.01000000"},{"id":"AFN","name":"Afghan Afghani","min_size":"0.01000000"},{"id":"ALL","name":"Albanian Lek","min_size":"0.01000000"},{"id":"AMD","name":"Armenian Dram","min_size":"0.01000000"},{"id":"ANG","name":"Netherlands Antillean Gulden","min_size":"0.01000000"},{"id":"AOA","name":"Angolan Kwanza","min_size":"0.01000000"},{"id":"ARS","name":"Argentine Peso","min_size":"0.01000000"},{"id":"AUD","name":"Australian Dollar","min_size":"0.01000000"},{"id":"AWG","name":"Aruban Florin","min_size":"0.01000000"},{"id":"AZN","name":"Azerbaijani Manat","min_size":"0.01000000"},{"id":"BAM","name":"Bosnia and Herzegovina Convertible Mark","min_size":"0.01000000"},{"id":"BBD","name":"Barbadian Dollar","min_size":"0.01000000"},{"id":"BDT","name":"Bangladeshi Taka","min_size":"0.01000000"},{"id":"BGN","name":"Bulgarian Lev","min_size":"0.01000000"},{"id":"BHD","name":"Bahraini Dinar","min_size":"0.00100000"},{"id":"BIF","name":"Burundian Franc","min_size":"1.00000000"},{"id":"BMD","name":"Bermudian Dollar","min_size":"0.01000000"},{"id":"BND","name":"Brunei Dollar","min_size":"0.01000000"},{"id":"BOB","name":"Bolivian Boliviano","min_size":"0.01000000"},{"id":"BRL","name":"Brazilian Real","min_size":"0.01000000"},{"id":"BSD","name":"Bahamian Dollar","min_size":"0.01000000"},{"id":"BTN","name":"Bhutanese Ngultrum","min_size":"0.01000000"},{"id":"BWP","name":"Botswana Pula","min_size":"0.01000000"},{"id":"BYN","name":"Belarusian Ruble","min_size":"0.01000000"},{"id":"BYR","name":"Belarusian Ruble","min_size":"1.00000000"},{"id":"BZD","name":"Belize Dollar","min_size":"0.01000000"},{"id":"CAD","name":"Canadian Dollar","min_size":"0.01000000"},{"id":"CDF","name":"Congolese Franc","min_size":"0.01000000"},{"id":"CHF","name":"Swiss Franc","min_size":"0.01000000"},{"id":"CLF","name":"Unidad de Fomento","min_size":"0.00010000"},{"id":"CLP","name":"Chilean Peso","min_size":"1.00000000"},{"id":"CNH","name":"Chinese Renminbi Yuan Offshore","min_size":"0.01000000"},{"id":"CNY","name":"Chinese Renminbi Yuan","min_size":"0.01000000"},{"id":"COP","name":"Colombian Peso","min_size":"0.01000000"},{"id":"CRC","name":"Costa Rican Colón","min_size":"0.01000000"},{"id":"CUC","name":"Cuban Convertible Peso","min_size":"0.01000000"},{"id":"CVE","name":"Cape Verdean Escudo","min_size":"0.01000000"},{"id":"CZK","name":"Czech Koruna","min_size":"0.01000000"},{"id":"DJF","name":"Djiboutian Franc","min_size":"1.00000000"},{"id":"DKK","name":"Danish Krone","min_size":"0.01000000"},{"id":"DOP","name":"Dominican Peso","min_size":"0.01000000"},{"id":"DZD","name":"Algerian Dinar","min_size":"0.01000000"},{"id":"EGP","name":"Egyptian Pound","min_size":"0.01000000"},{"id":"ETB","name":"Ethiopian Birr","min_size":"0.01000000"},{"id":"EUR","name":"Euro","min_size":"0.01000000"},{"id":"FJD","name":"Fijian Dollar","min_size":"0.01000000"},{"id":"FKP","name":"Falkland Pound","min_size":"0.01000000"},{"id":"GBP","name":"British Pound","min_size":"0.01000000"},{"id":"GBX","name":"British Penny","min_size":"1.00000000"},{"id":"GEL","name":"Georgian Lari","min_size":"0.01000000"},{"id":"GGP","name":"Guernsey Pound","min_size":"0.01000000"},{"id":"GHS","name":"Ghanaian Cedi","min_size":"0.01000000"},{"id":"GIP","name":"Gibraltar

Pound","min_size":"0.01000000"},{"id":"GMD","name":"Gambian
Dalasi","min_size":"0.01000000"},{"id":"GNF","name":"Guinean
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Dollar","min_size":"0.01000000"},{"id":"HKD","name":"Hong Kong
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Forint","min_size":"1.00000000"},{"id":"IDR","name":"Indonesian
Rupiah","min_size":"0.01000000"},{"id":"ILS","name":"Israeli New
Sheqel","min_size":"0.01000000"},{"id":"IMP","name":"Isle of Man
Pound","min_size":"0.01000000"},{"id":"INR","name":"Indian
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Dollar","min_size":"0.01000000"},{"id":"JOD","name":"Jordanian
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Shilling","min_size":"0.01000000"},{"id":"KGS","name":"Kyrgyzstani
Som","min_size":"0.01000000"},{"id":"KHR","name":"Cambodian
Riel","min_size":"0.01000000"},{"id":"KMF","name":"Comorian
Franc","min_size":"1.00000000"},{"id":"KRW","name":"South Korean
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Sol","min_size":"0.01000000"},{"id":"PGK","name":"Papua New Guinean
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Dinar","min_size":"0.01000000"},{"id":"RUB","name":"Russian
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Franc","min_size":"1.00000000"},{"id":"SAR","name":"Saudi
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Rupee","min_size":"0.01000000"},{"id":"SEK","name":"Swedish
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Dollar","min_size":"0.01000000"},{"id":"SSP","name":"South Sudanese
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Colón","min_size":"0.01000000"},{"id":"SZL","name":"Swazi
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Pa‘anga","min_size":"0.01000000"},{"id":"TRY","name":"Turkish
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purposes","min_size":"1.00000000"},{"id":"YER","name":"Yemeni
Rial","min_size":"0.01000000"},{"id":"ZAR","name":"South African
Rand","min_size":"0.01000000"},{"id":"ZMW","name":"Zambian
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Kwacha","min_size":"0.01000000"},{"id":"ZWD","name":"Zimbabwean
Dollar","min_size":"0.01000000"},{"id":"ZWL","name":"Zimbabwean
Dollar","min_size":"0.01000000"}]}